STATEMENT OF WORK

1. Objective/Requirements

The Autonomous Science and Technology for Responsive Adaptability (ASTRA) project goals are to develop and test an artificial intelligence (AI) agent designed specifically for onboard complex decision-making that can adapt to updated mission priorities. To achieve these goals within the fiscal year timeline, ASTRA requires a hyperspectral imager that has a low enough Size, Weight, and Power (SWAP) to be integrated onto and used on a variety of spaceflight-relevant robotic vehicles (e.g., uncrewed aerial systems, ballons, small rovers). Specifically, ASTRA requires a payload that is less than 500 g. ASTRA also requires a hyperspectral imager that has greater than 30-degree field of view, a mosaic bandpass filter, and at least 5 nm spectral sampling resolution in order to perform the required tests.

2. Characteristics, Scope, and Specs

The imager/spectrometer shall perform the following duties and specifications:

- Perform hyperspectral imaging and spectral graphing over the wavelength range of 350 – 1000 nm.
  - Perform imaging and spectral sampling in laboratory and outdoor environments.
- Perform spectral sampling at a resolution of 5 nm.
- Perform bandpass filtering using a mosaic.
- Capture images and spectra over a 30 degree field of view.
- Maintain a standalone mass of < 500 g.

LINE	PART NO.	DESCRIPTION	QTY.
1	X20	**ULTRIS X20 Light Field Camera** • Wavelength range: 350 - 1000 nm • Sensor size: 20MP • Spectral Bands: 164 • Spectral Sampling: 4nm • FWHM: Constant 10 nm • Spatial Resolution: 410 x 410px (168,100 spectra/cube) • Data Depth: 12bit • Max. Frame Rate: 8 Hz • Data Link: GigE • Field 0f View (FOV): typ. 35° • Weight: 350g • Size: 57x60x60mm	1
2	Flight Lite U	**Flight Lite Package** • Cubert CUVIS HSI Software • Cubert Touch (GUI) & Core for Windows • Cubert CubeExport • Cubert SDK (Python, C++, C#) • Radiometric Calibration • Flight Recorder Pokini F2 (Mini PC) for customized / mobile solutions • USG SiRFIV USB GPS Receiver • Camera Accessory Package (cables, protection case, power supply, reference target)	1

3. Place of Performance

Code 699, Planetary Environments Laboratory
NASA Goddard Space Flight Center
Greenbelt, MD
[redacted]

4. Period of Performance

ASTRA testing will begin during June 2025 and is projected to continue through August 2026 with opportunities for follow-on activities.

Estimated delivery lead-time is 8/12 weeks ARO.